Transferring MaizeGDB Data with Globus

MaizeGDB Public Collection · SCINet – JUNO

DRAFT

maizegdb.org/download
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Globus is a file transfer service for research data. It runs transfers in the background on servers rather than in your browser, restarts them by itself after a network failure, verifies every file by checksum, and emails you when the job finishes. That makes it the right tool for whole-genome sets, variant matrices, and anything else large enough that an interrupted browser download would cost you an afternoon.

COLLECTION	MaizeGDB Public Collection (SCINet – JUNO)
ENDPOINT	`g-3126c1.67741c.a567.data.globus.org`
TYPE	Guest collection on SCINet-Juno
CONTACT	`carson.andorf@scinet.usda.gov`

1 Before you start

You need a Globus account. It is free, and most researchers already have access through their institution.

1. Go to `app.globus.org` and choose **Log In**.
2. Search for your university, agency, or lab in the organization list and sign in with those credentials. If your organization is not listed, create a Globus ID from the same screen instead.

> **If the collection refuses you after login:** the MaizeGDB collection is intended for open, read-only public access, but access is administered per identity. Email the contact address above with the identity you signed in with and it can be checked.

2 Choose where the data should land

Globus copies between two *collections*. MaizeGDB is the source; you supply the destination. Which one you use depends on where you want the files.

DESTINATION	WHAT TO USE
University or agency cluster	Almost always already a Globus collection. Search your institution's name in the File Manager; ask your research computing group for the exact collection name if several appear.
Your own laptop or lab workstation	**Globus Connect Personal** — a small free application that turns the machine into a collection. Install it from `globus.org/globus-connect-personal`, sign in, and give the collection a name.
Cloud storage	Google Drive, Amazon S3, and Box connectors exist but are enabled per institution. Check with your research computing group first.

③ Transfer in the web interface

The point-and-click route. Nothing to install beyond the destination collection.

1. Open `app.globus.org/file-manager` and log in.
2. Click **Collection** in the left panel and paste the endpoint `g-3126c1.67741c.a567.data.globus.org`, or search for *MaizeGDB Public Collection*.
3. Browse to the files you want and select them. Selecting a folder transfers everything beneath it.
4. Switch to the right panel with **Transfer or Sync to…** and choose your destination collection and folder.
5. Click **Start**. The transfer now runs without you — you can close the browser. Globus emails you when it completes.

> **Transfer Settings** is worth one look before a large job. *Sync — only transfer new or changed files* makes a repeated transfer resumable and cheap, which is what you want when refreshing a local copy of a genome release.

❹ Transfer from the command line

The command line route suits batch jobs, scheduled refreshes, and anything that belongs in a pipeline script.

Install and sign in

```
# Install the CLI (pip or conda both work)
pip install globus-cli

# Authenticate once; the session persists
globus login
```

Find the collection and look around

```
# MaizeGDB's endpoint ID, used as SOURCE below
export SOURCE=g-3126c1.67741c.a567.data.globus.org

# List the top level of the collection
globus ls $SOURCE:/

# Find your own destination collection's ID
globus endpoint search --filter-scope my-endpoints
```

Start a transfer

```
# One directory, recursively, to your endpoint
globus transfer $SOURCE:/genomes/ <YOUR_ENDPOINT_ID>:/scratch/maize/ \
  --recursive --label "MaizeGDB genome sync"

# Watch it, or let it run and check back later
globus task list
globus task wait <TASK_ID>
```

> **Paths are collection-relative.** `/genomes/` here is a path inside the MaizeGDB collection, not a path on your own machine, and it is not the same as a URL on the download server. Run `globus ls $SOURCE:/` first and copy the directory name you see rather than assuming one.

5 When Globus is not the right tool

Globus earns its setup cost on large or repeated transfers. For a handful of files, the other two MaizeGDB sources are faster to reach:

- **MaizeGDB FTP** — `ftp.maizegdb.org`, served over HTTPS. Genome assemblies and annotations are under `/downloads/`, also reachable as `download.maizegdb.org`. Every file has a stable URL, so `wget` and `curl` work without an account. The guided genome file finder at `maizegdb.org/download` will build those commands for you.
- **MaizeGDB Box** — project collections, machine learning model weights, and bundled genome-browser track archives.

6 Getting help

QUESTION	WHERE TO GO
What is in a file, or which release to use	MaizeGDB contact form at `maizegdb.org/contact`
Access denied to the collection after login	`carson.andorf@scinet.usda.gov`, quoting the identity you used
Globus accounts, connectors, and endpoints at your institution	Your local research computing group
Globus itself — errors, CLI reference	`docs.globus.org`